Requirement provenance — where each of the 82 features came from

A mark means the source's published documentation evidenced this requirement. It does NOT mean the product has the feature, and a blank does NOT mean it lacks one.

The research behind the gap matrix was a handful of published pages per vendor — enough to establish what the market expects a key management system to do, nowhere near enough to compare products feature by feature. A capability comparison would need a hands-on evaluation of each product, and this is not one.

THA	Thales CipherTrust Manager	**FOR**	Fortanix Data Security Manager
ENT	Entrust KeyControl	**UTI**	Utimaco Enterprise Secure Key Manager
BLO	Bloombase KeyCastle	**COS**	Cosmian / Eviden KMS
SEC	Securosys CyberVault KMS	**VLT**	HashiCorp Vault
CLD	AWS KMS / Azure Managed HSM / Google Cloud KMS	**STD**	NIST SP 800-57, SP 800-152, OASIS KMIP 2.1 and 3.0
CAT	Category baseline — no source named it		

CAT is the honest residue: requirements no source in the research named, present because they are ordinary practice for a networked service that handles secrets. It never sits beside another code — the generator refuses to build if it does — so the CAT column counts exactly the requirements nothing evidenced, and those rows are marked in amber rather than left to hide among the others.

Marks are given strictly, and that costs blank cells. A vendor whose documentation describes audit logging has evidenced an audit trail, not a tamper-evident one, so it leaves the tamper-evidence row blank. Read a blank as “not seen in what was read”, never as “absent from the product”.

Key lifecycle and cryptography

Requirement	THA	FOR	ENT	UTI	BLO	COS	SEC	VLT	CLD	STD	CAT	Status
HSM-backed storage, keys never leave	●	●	●	●	●	●	●	·	●	●	·	Covered
Full lifecycle state machine	●	·	●	·	●	●	●	·	·	●	·	Covered
Cryptoperiod enforcement	·	·	●	·	·	·	·	·	●	●	·	Covered
Automatic rotation	●	·	●	●	·	●	●	·	●	·	·	Partial
Key lineage and versioning	●	·	·	·	·	·	·	●	●	·	·	Covered
Algorithm breadth	·	·	·	·	·	●	●	●	·	●	·	Partial
Post-quantum (ML-KEM, ML-DSA, SLH-DSA)	·	·	·	·	·	●	●	·	·	●	·	Gap
Hardware RNG	·	·	·	●	●	·	·	·	·	●	·	Covered
Key derivation	·	·	·	·	·	·	·	●	·	●	·	Covered
Key wrapping / secure import-export	●	·	·	·	●	·	·	●	●	●	·	Covered
Split knowledge / M-of-N shares	·	●	·	●	·	·	·	●	·	●	·	Partial
Formal key ceremony	·	·	·	·	·	·	·	●	·	●	·	Gap
Bulk and batch operations	·	·	·	·	·	·	·	·	·	●	·	Partial
Verifiable destruction	●	·	●	·	·	·	·	·	·	●	·	Covered
Key-bound usage policy enforced in hardware	·	·	·	·	·	·	●	·	·	·	·	Gap

Protocol and ecosystem integration

Requirement	THA	FOR	ENT	UTI	BLO	COS	SEC	VLT	CLD	STD	CAT	Status
KMIP server	●	·	●	●	●	●	●	●	·	●	·	Covered
KMIP interoperability certification	·	·	·	·	●	·	·	·	·	●	·	Gap
KMIP 3.0	·	·	·	·	·	·	·	·	·	●	·	Gap
PKCS#11 provider for applications	·	·	·	●	●	●	●	·	·	·	·	Gap
REST / JSON API	●	●	·	●	·	●	●	●	●	·	·	Gap
Cloud BYOK (AWS, Azure, GCP)	●	●	·	·	·	·	·	·	●	·	·	Gap
Cloud EKM / XKS / hold-your-own-key	·	●	·	·	·	·	●	·	●	·	·	Gap
Database TDE integration	●	·	●	·	·	●	●	·	·	·	·	Partial
Storage, backup and VM integrations	·	·	●	·	●	·	●	·	·	·	·	Partial
Secrets management	●	●	●	·	·	·	·	●	·	·	·	Partial
Tokenization / format-preserving encryption	●	●	·	·	·	●	·	·	·	·	·	Gap
Certificate lifecycle	·	·	·	·	●	●	●	·	·	·	·	Partial
Encryption as a service (data-plane API)	·	·	·	·	·	●	·	●	●	·	·	Partial
KMIP JSON and XML encodings	·	·	·	·	·	●	·	·	·	●	·	Gap
OpenAPI specification and generated clients	·	·	·	·	·	●	·	·	●	·	·	Gap

Identity and access control

Requirement	THA	FOR	ENT	UTI	BLO	COS	SEC	VLT	CLD	STD	CAT	Status
Per-identity authentication	●	·	·	●	·	●	·	●	·	·	·	Covered
Enterprise IdP — LDAP/AD, SAML, OIDC	·	·	·	●	·	●	·	●	·	·	·	Gap

mTLS certificate identity	·	·	·	·	·	●	·	●	·	·	·	Covered
API keys / service accounts	·	·	·	·	·	●	·	●	●	·	·	Gap
Role-based access control	●	·	·	●	·	·	●	●	·	·	·	Covered
Groups	·	·	·	·	·	·	·	●	·	·	·	Covered
Per-object access grants	●	·	·	·	·	●	·	·	·	·	·	Covered
Attribute or policy-based access	●	●	●	·	·	●	●	·	·	·	·	Gap
Dual control / quorum approval	·	●	·	●	·	·	●	●	·	·	·	Covered
Separation of duties	·	·	·	·	·	·	·	·	·	●	·	Partial
Multi-tenancy and namespaces	●	·	●	·	·	·	●	●	·	·	·	Gap
Per-tenant quotas	·	·	·	·	·	·	·	●	·	·	·	Gap
Connection and rate limiting	·	·	·	·	·	·	·	●	·	·	·	Gap

Audit, compliance and assurance

Requirement	THA	FOR	ENT	UTI	BLO	COS	SEC	VLT	CLD	STD	CAT	Status
Tamper-evident audit log	●	·	·	●	·	·	·	·	·	●	·	Covered
External audit anchoring	·	·	·	·	·	·	·	·	·	·	●	Gap
SIEM integration	·	·	·	●	●	·	·	●	·	·	·	Gap
Retention and archival	·	·	·	·	●	·	·	·	·	●	·	Covered
FIPS 140-2/3 validated HSM	·	●	●	·	●	●	●	·	●	·	·	Gap
Common Criteria / eIDAS	·	·	●	·	·	·	●	·	·	·	·	Gap
Compliance reporting	●	·	●	·	·	·	●	·	·	·	·	Gap
Key inventory and discovery	●	·	●	·	·	·	●	·	·	·	·	Partial
Cryptoperiod policy alignment	·	·	·	·	·	·	·	·	·	●	·	Covered
Crypto-agility reporting	·	·	·	·	·	·	●	·	·	·	·	Partial
Certificate discovery and expiry monitoring	·	·	·	·	·	·	●	·	·	·	·	Gap

Availability, scale and recovery

Requirement	THA	FOR	ENT	UTI	BLO	COS	SEC	VLT	CLD	STD	CAT	Status
HA clustering	●	·	·	●	●	●	·	●	·	·	·	Gap
Multi-site replication and DR	·	·	●	●	·	·	·	●	·	●	·	Gap
HSM failover and pooling	·	·	·	●	●	·	·	·	·	·	·	Gap
Horizontal throughput	●	·	·	·	·	●	·	●	·	·	·	Partial
Enterprise database backend	·	·	·	·	·	●	·	●	·	·	·	Gap
Scheduled and offsite backup	●	·	●	·	·	·	·	·	·	●	·	Partial
Restore that proves itself	·	·	·	·	·	·	·	·	·	●	·	Covered
Zero-downtime certificate rotation	·	·	·	·	·	·	·	·	·	·	●	Covered
Rolling upgrades	·	·	·	·	·	·	·	·	·	·	●	Covered
Kubernetes / container deployment	●	·	·	·	·	●	●	·	·	·	·	Partial
Confidential computing deployment	·	·	·	·	·	●	·	·	·	·	·	Gap

Operations and observability

Requirement	THA	FOR	ENT	UTI	BLO	COS	SEC	VLT	CLD	STD	CAT	Status
Health and readiness probes	·	·	·	·	·	·	·	·	·	·	●	Covered
Metrics	·	·	·	·	·	·	·	●	·	·	·	Covered
Structured logging	·	·	·	·	·	·	·	·	·	·	●	Covered
Configuration-driven deployment	·	·	·	·	·	·	·	·	·	·	●	Covered
Administrative CLI	·	·	·	·	·	●	·	●	·	·	·	Covered
Web console	●	●	●	·	●	●	●	·	·	·	·	Gap
Self-service developer portal	●	·	·	·	·	·	·	·	·	·	·	Gap
Alerting	·	·	·	·	●	·	●	·	·	·	·	Partial
Distributed tracing	·	·	·	·	·	●	·	·	·	·	·	Gap

Platform hardening

Requirement	THA	FOR	ENT	UTI	BLO	COS	SEC	VLT	CLD	STD	CAT	Status
TLS enforced by default	·	·	·	·	·	·	·	·	·	·	●	Covered
Metadata encrypted at rest	·	·	·	·	·	●	·	●	·	●	·	Covered
Master key rotation	·	·	·	·	·	·	·	●	·	●	·	Covered
Non-extractable key enforcement	·	·	·	·	·	·	●	·	●	●	·	Covered
Request bounds and DoS resistance	·	·	·	·	·	·	·	·	·	·	●	Partial
Secrets kept out of configuration	·	·	·	·	·	·	·	·	·	·	●	Covered
Hardened service	·	·	·	·	·	·	·	·	·	·	●	Covered
Sealed startup with quorum unseal	·	·	·	·	·	·	·	●	·	·	·	Partial

What the columns add up to

Read this as a description of the research, not of the market. A source with few marks is one whose documentation I read less of, not a product with fewer features.

Source	Requirements evidenced	Share of 82	Kind
THA — Thales CipherTrust Manager	26	31 %	Vendor documentation
FOR — Fortanix Data Security Manager	11	13 %	Vendor documentation
ENT — Entrust KeyControl	18	21 %	Vendor documentation
UTI — Utimaco Enterprise Secure Key Manager	16	19 %	Vendor documentation
BLO — Bloombase KeyCastle	16	19 %	Vendor documentation
COS — Cosmian / Eviden KMS	30	36 %	Vendor documentation
SEC — Securosys CyberVault KMS	27	32 %	Vendor documentation
VLT — HashiCorp Vault	30	36 %	Vendor documentation
CLD — AWS KMS / Azure Managed HSM / Google Cloud KMS	13	15 %	Vendor documentation
STD — NIST SP 800-57, SP 800-152, OASIS KMIP 2.1 and 3.0	26	31 %	Standard
CAT — Category baseline — no source named it	10	12 %	Category baseline

Requirements carried only by the category baseline: 10 of 82

These are the rows to challenge first if the requirement list is ever disputed. Each is defensible as ordinary practice, but none was named by a vendor or a standard in the research behind the matrix.

- External audit anchoring
- Zero-downtime certificate rotation
- Rolling upgrades
- Health and readiness probes
- Structured logging
- Configuration-driven deployment
- TLS enforced by default
- Request bounds and DoS resistance
- Secrets kept out of configuration
- Hardened service

What would make this stronger

- A hands-on evaluation of one or two products, which would turn this provenance table into a genuine capability comparison.
- The vendors' administration guides rather than their marketing pages — the former say what a product does, the latter what it advertises.
- An analyst grid, which would add an independent view of the category rather than each vendor's own.
- Direct access to four of the vendors' documentation sites. Bloombase, Cosmian, Securosys and HashiCorp were researched through indexed summaries of their pages because the network policy where this document is built blocks all four domains — so their columns are thinner than the reading would otherwise support.